IEEE Education Society
SEC IEEE SBC

<h1>EDSPARK</h1>

# Web Architecture 101: Your First Site





<h4> Presented By:</h4>

 Rifaath Fathimah 

 Lakshan M 

 Mythili R 

# HTML INTRO

IEEE Education Society
SEC IEEE SBC

- HTML = HyperText Markup Language

**Think of a website as a building:**

| 🏗️ Website | 🏠 Building |
|---|---|
| HTML | Structure |
| CSS | Paint & Decoration |
| JavaScript | Electricity & Controls |

HTML 5

File Edit Selection View Go Run Terminal Help

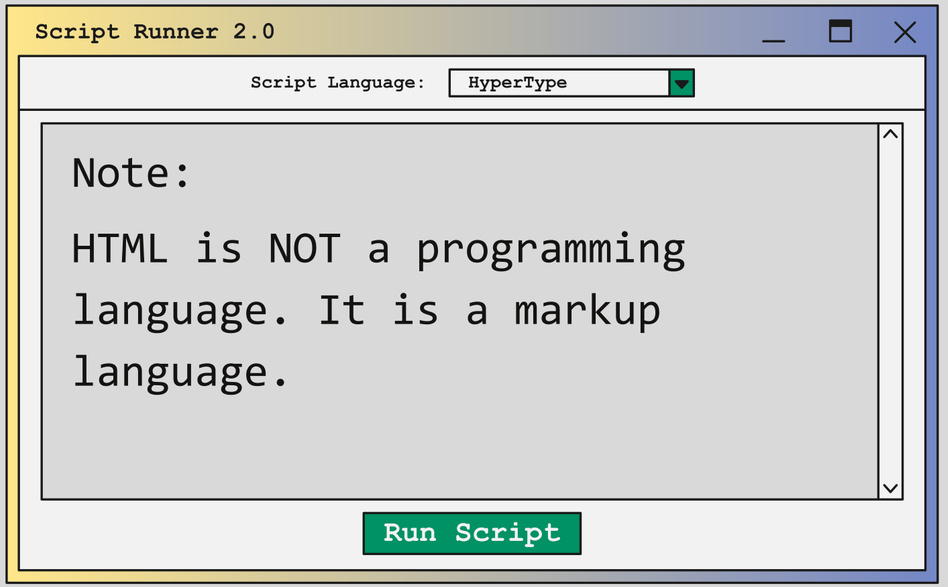
Intro ×

# HTML INTRO

IEEE Education Society
SEC IEEE SBC

HTML defines:

- 📝 Text
- 🖼️ Images
- 🔗 Links
- 📋 Lists
- 📦 Sections
- ✏️ Input
- 📝 Forms
- 🔘 Buttons




HTML5

File Edit Selection View Go Run Terminal Help

Work ×

# How HTML Works

- Most HTML elements have:

```
<tag> Content </tag>
```

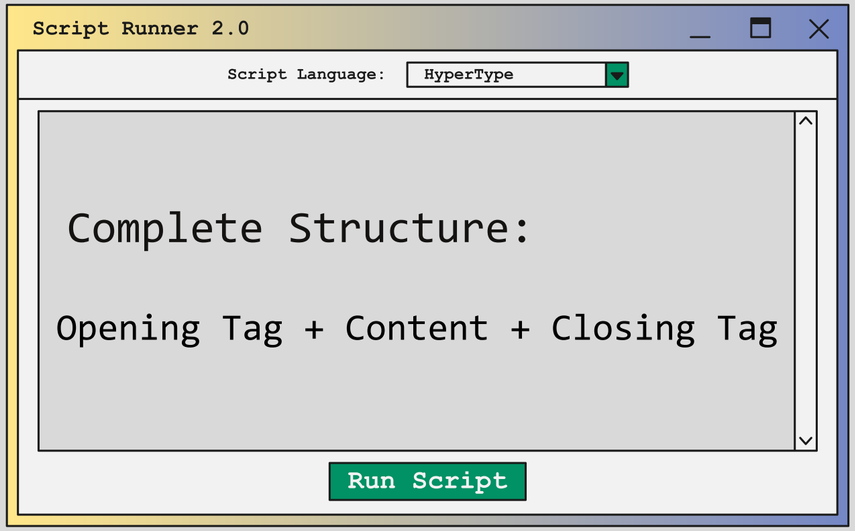
- Example:

```
<h1> Welcome to EdSpark! </h1>
```

Opening Tag

Content

Closing Tag




IEEE Education Society
SEC IEEE SBC

HTML

File Edit Selection View Go Run Terminal Help

The Basic HTML Structure ×

# The Basic HTML Structure

```
<!DOCTYPE html>

<html>
<head>
    <title>My First Website</title>
</head>
                                    <!-- This is a comment -->
<body>

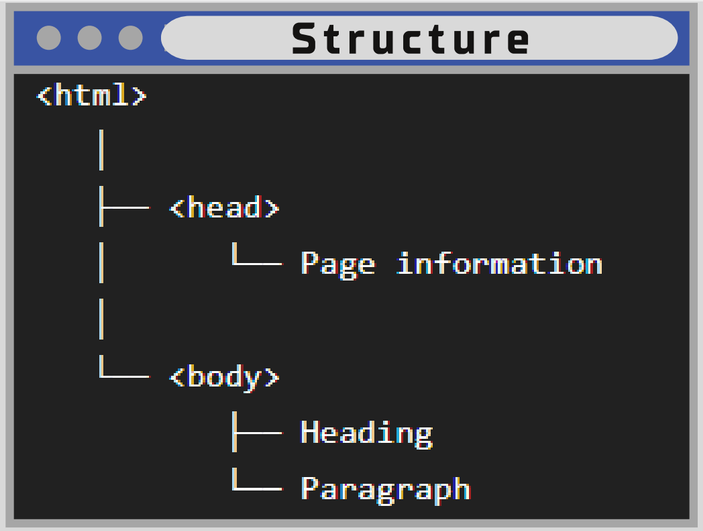
    <h1>Hello, EdSpark!</h1>
    <p>This is my first website.</p>

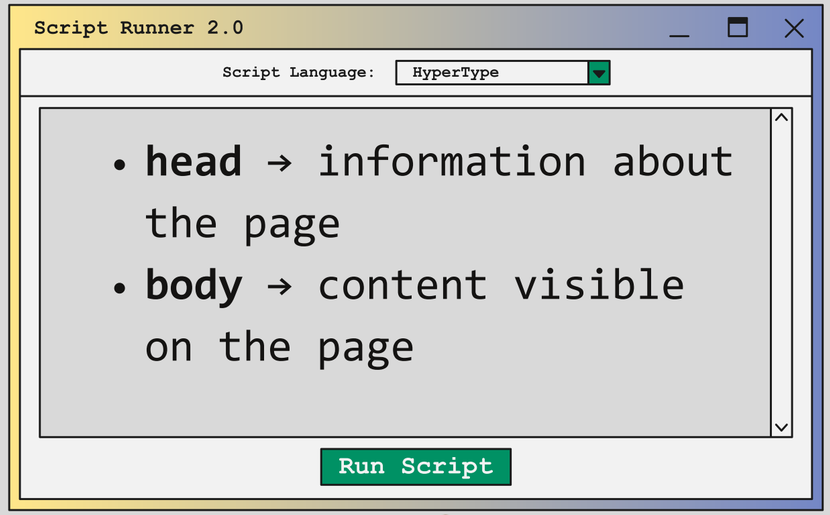
</body>
</html>
```

**<!-- This is a comment -->**

- Comments are **not displayed on the webpage**

- Why use comments?
- 🧠 Explain your code
- 🔍 Organize sections
- 🛠️ Make debugging easier




IEEE Education Society
SEC IEEE SBC

File Edit Selection View Go Run Terminal Help

Head ×

# The <head> Section

IEEE Education Society
SEC IEEE SBC

- The <head> contains information about the webpage.
- Example code:

```
<head>
    <meta charset="UTF-8">
    <meta name="viewport"content="width=device-width, initial-scale=1.0">
    <title>EdSpark</title>
</head>
```

| Element | Purpose |
|---|---|
| <title> | Browser tab title |
| <meta> | Page information |
| <link> | Connect CSS / resources |
| <style> | Internal CSS |
| <script> | JavaScript |

HTML5

File Edit Selection View Go Run Terminal Help

Semantic & Non-Semnatic Tags ×

# SEMANTIC & NON-SEMANTIC TAGS

IEEE Education Society
SEC IEEE SBC

## Semantic vs. Non-Semantic HTML Tags: Direct Comparison

| Feature | Semantic Elements | Non-Semantic Elements |
|---|---|---|
| **Definition** | Tags that carry explicit contextual meaning. | Generic structural containers with no inherent meaning. |
| **Core Examples** | `<header>`, `<nav>`, `<main>`, `<article>`, `<section>`, `<footer>` | `<div>`, `<span>` |
| **SEO Impact** | **High:** Helps search engine bots index and rank pages efficiently. | **Neutral:** Offers no direct context to crawler engines. |
| **Accessibility** | **Excellent:** Crucial for screen readers used by visually impaired users. | **Poor:** Treats components as generic blocks without assistive flags. |
| **Code Readability** | Self-explanatory codebase that simplifies collaboration. | Heavily relies on classes or IDs (`<div id="nav">`) to make sense. |

HTML 5

File Edit Selection View Go Run Terminal Help

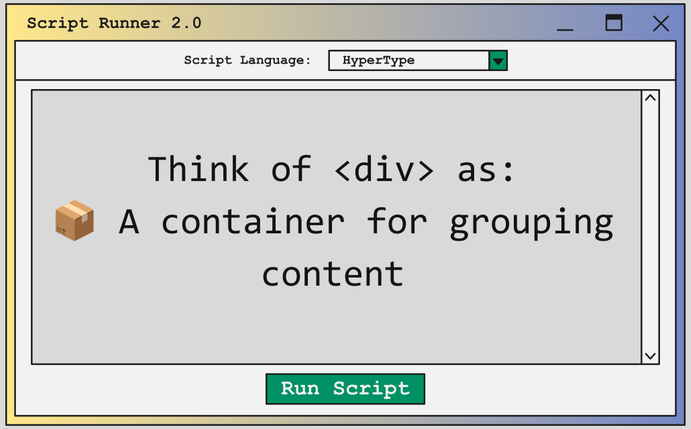
Containers ×

# HTML CONTAINERS

- Two extremely common elements:

```
<div>
    ...
</div>
```

```
<span>
    ...
</span>
```




|  |  |
|---|---|
| **Block-level container** | **Inline container** |
| Groups larger sections | Groups small pieces of text |
| Used for layouts/sections | Used inside text |
| Can contain many elements | Usually short content |

IEEE Education Society
SEC IEEE SBC

File Edit Selection View Go Run Terminal Help

Headings & Paragraphs ×

# HEADINGS & PARAGRAPHS

IEEE Education Society
SEC IEEE SBC

- Adds Text to Your Website
- Headings

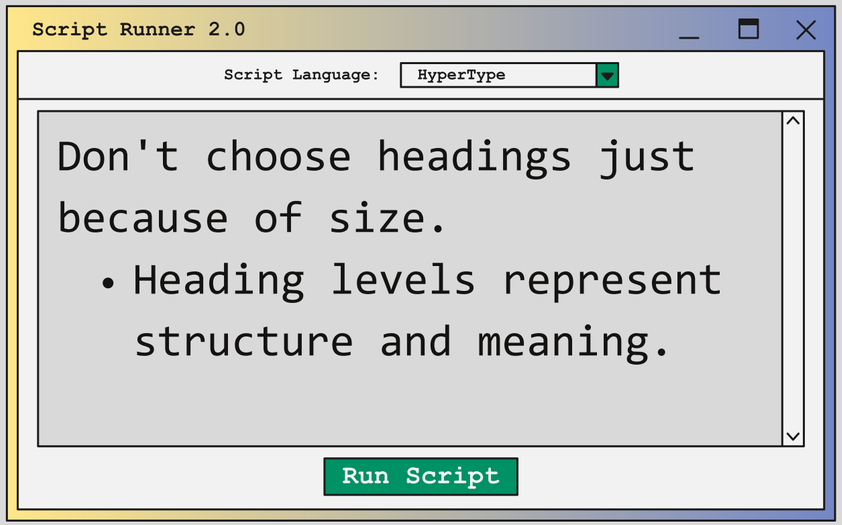
```
<h1>Main Heading</h1>
<h2>Section Heading</h2>
<h3>Subsection</h3>
```

Paragraph

```
<p>
    Welcome to my first website!
</p>
```

Heading hierarchy:

```
<h1>Heading 1</h1>
<h2>Heading 2</h2>
<h3>Heading 3</h3>
<h4>Heading 4</h4>
<h5>Heading 5</h5>
<h6>Heading 6</h6>
```



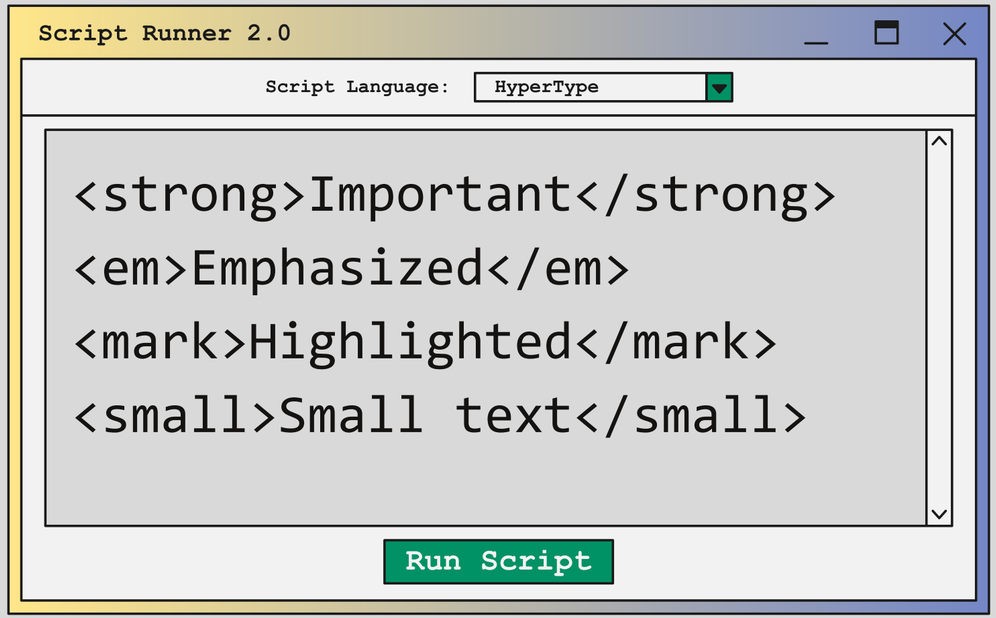
File Edit Selection View Go Run Terminal Help

TEXT FORMATTING

# TEXT FORMATTING

IEEE Education Society
SEC IEEE SBC

- Make Content Meaningful!




Example:

```
<p>
    Learn <strong>HTML</strong>
    and build your
    <em>first website!</em>
</p>
```

HTML5

File Edit Selection View Go Run Terminal Help

Attributes ×

# HTML ATTRIBUTES

IEEE Education Society
SEC IEEE SBC

- Adds Extra Information
  - HTML elements can have attributes.

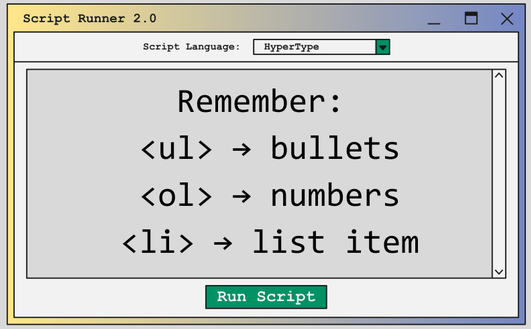
```
<tag attribute="value">
```

| Attribute | Used for |
|---|---|
| id | Unique identification |
| class | Grouping / styling |
| href | Link destination |
| src | Resource location |
| alt | Image description |
| title | Extra information |

File Edit Selection View Go Run Terminal Help

Lists ×

# LISTS




- HTML provides two common types of lists:

| Type | Tag | Use |
|---|---|---|
| Unordered | <ul> | Bullet points |
| Ordered | <ol> | Numbered steps |

```
<h3>My Skills</h3>

<ul>
  <li>HTML</li>
  <li>CSS</li>
  <li>JavaScript</li>
</ul>
```

```
<h3>How to Build a Website</h3>

<ol>
  <li>Plan</li>
  <li>Write HTML</li>
  <li>Add CSS</li>
  <li>Add JavaScript</li>
</ol>
```

IEEE Education Society
SEC IEEE SBC

# INPUT ELEMENT

IEEE Education Society
SEC IEEE SBC

| Type | Description |
| --- | --- |
| <input type="text"> | Displays a single-line text input field |
| <input type="email"> | Displays a single-line text input field for email |
| <input type="radio"> | Displays a radio button (for selecting one of many choices) |
| <input type="checkbox"> | Displays a checkbox (for selecting zero or more of many choices) |
| <input type="submit"> | Display a submit button (for submitting the form) |
| <input type="button"> | Displays a clickable button |
| <input type="password"> | Displays a text box to enter password |

File Edit Selection View Go Run Terminal Help

Tables ×

# TABLE

IEEE Education Society
SEC IEEE SBC

- Tables are useful for displaying structured information.

| Tag | Purpose |
| --- | --- |
| <table> | Creates the table |
| <tr> | Table row |
| <th> | Table heading |
| <td> | Table data |

HTML 5

File Edit Selection View Go Run Terminal Help

Tables ×

# TABLE

IEEE Education Society
SEC IEEE SBC

- Example:

```
<table>
  <tr>
    <th>Name</th>
    <th>Score</th>
  </tr>

  <tr>
    <td>Arun</td>
    <td>95</td>
  </tr>

  <tr>
    <td>Priya</td>
    <td>90</td>
  </tr>
</table>
```

HTML 5

File Edit Selection View Go Run Terminal Help

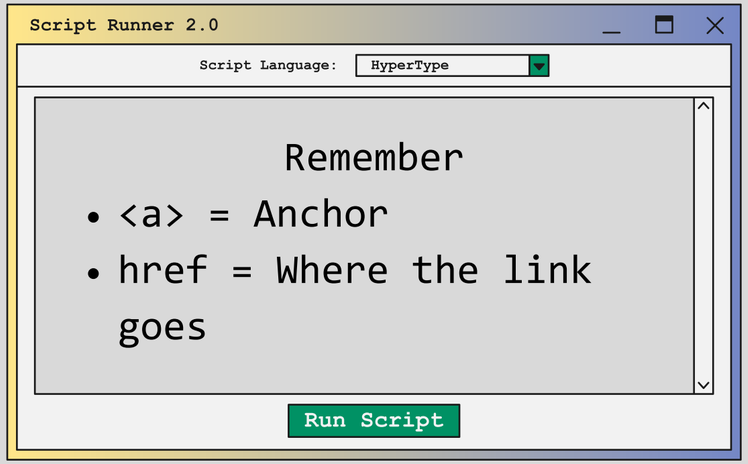
Anchor ×

# ANCHOR TAG

- Create Links with <a>

  The anchor element is used to create links.

```
<a href="https://www.google.com">
    Visit Google
</a>
```




To open in a new tab:

```
<a href="https://github.com"
   target="_blank">
   My GitHub
</a>
```

IEEE Education Society
SEC IEEE SBC

File Edit Selection View Go Run Terminal Help

Image Video & Audio ×

# IMAGE, VIDEO & AUDIO

IEEE Education Society
SEC IEEE SBC

Adds Media to Your Website

HTML allows you to include different types of media.

- Images:

```
<img src="profile.jpg" alt="Profile photo">
```

- Video:

```
<video controls>
<source src="intro.mp4" type="video/mp4">
</video>
```

- Audio:

```
<audio controls>
<source src="music.mp3" type="audio/mpeg">
</audio>
```

HTML5

File Edit Selection View Go Run Terminal Help

Challenge ×

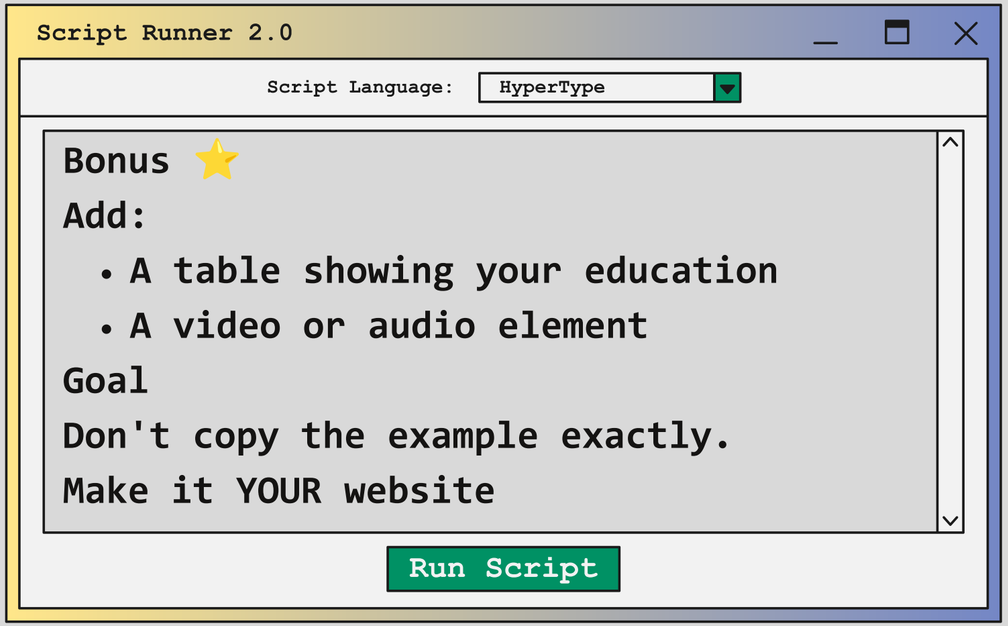
# HTML CHALLENGE

**Build Your Personal Profile**

Your Challenge

Create a simple webpage containing:

1. Heading : Your Name
2. Introduction: A short paragraph about yourself
3. Skills
   - HTML
   - CSS
   - JavaScript
   - Your own skill
4. Profile Image
5. Links
   - GitHub
   - LinkedIn
6. Contact
   - Email input
   - Submit button




IEEE Education Society
SEC IEEE SBC

File Edit Selection View Go Run Terminal Help

Your Website Starts Here ×

# YOUR WEBSITE STARTS HERE

IEEE Education Society
SEC IEEE SBC

- HTML is just the beginning.
- You've learned how to build the structure of a webpage
- Your Challenge
  - Take an idea.
  - Turn it into a webpage.
  - Make it yours.
- Your first website doesn't have to be perfect.
  It just has to be your first.

HTML5

IEEE Education Society
SEC IEEE SBC

<h1> THANK YOU FOR LISTENING </h1>

EdSpark — Ignite Your First Website
Learn → Build → Experiment → Create